

Online Supplement

Predicting Space-Use with Continuous-Time Markov Chain Movement Models

Devin S. Johnson^{1,*}, Michaela A. Kratofil^{2,3}, Janelle J. Gardner¹,

Robin W. Baird³, and Amanda L. Bradford¹

¹Pacific Islands Fisheries Science Center, National Marine Fisheries
Service, NOAA, Honolulu, Hawaii, USA

²Marine Mammal Institute, Department of Fisheries, Wildlife, and
Conservation Sciences, Oregon State University, Newport, Oregon,
USA

³Cascadia Research Collective, Olympia, Washington, USA

*Corresponding author: devin.johnson@noaa.gov

August 11, 2026

S1 Proof of closed form CTMC limiting distribution

Using the Metropolis-Hastings (MH) algorithm we can show, by construction, that the limiting distribution of the CTMC we are considering has the limiting distribution stated in Section 3.1. Recall that the limiting distribution of a CTMC is $\mathbf{u}$ with i th element $u_i = \pi_i/\Lambda_i$ where $\boldsymbol{\pi}$ is the limiting distribution of the embedded chain, $\mathcal{G}$. So, we can use the theoretical results of the MH algorithm to show that the discrete-time chain, $\mathcal{G}$, has the appropriate limiting distribution.

In general, the MH algorithm is designed to sample from a distribution $c^{-1}\pi(x)$ when the proportionality constant is unknown. It accomplishes this feat by constructing a Markov process $q(X_t|X_{t-1})$ to move from state X_{t-1} to X_t such that, asymptotically, $X_t \sim c^{-1}\pi(x)$ as $t \rightarrow \infty$. Full details of the MH algorithm can be found in Chib and Greenberg (1995), but the basic version proceeds as follows for desired limiting distribution, $c^{-1}\pi(x)$,

1. For current state $X_{t-1} = x$ propose $X_t = x'$ from probability distribution $q(x'|x)$.

2. With probability

$$R_t = \min \left\{ \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}, 1 \right\}$$

set $X_t = x'$ else, set $X_t = x$.

3. Repeat.

The acceptance step of the MH algorithm is the key feature that ensures, under certain general regularity conditions, that the distribution of $X_t \sim c^{-1}\pi(x)$ as $t \rightarrow \infty$. The Markov process $q(\cdot|\cdot)$ need not necessarily have a limiting distribution, but the acceptance step will produce a chain with limiting distribution $c^{-1}\pi(x)$. If $R_t \geq 1$ for all t , then no matter what

state is proposed by $q(\cdot|\cdot)$, it is accepted, thus, it is the same as simply drawing a realization of the $q(\cdot|\cdot)$ process. In that case, the limiting distribution of $q(\cdot|\cdot)$ is $c^{-1}\pi(x)$ by construction.

We now return to the explicit UD CTMC model proposed in the paper to prove the resulting limiting distribution is as stated. Recall the proposed CTMC has movement rate function of the form,

$$\lambda_{ij}(t) = \exp\{a_i + b_j + (c_j - c_i) + d_{ij}\}.$$

The embedded Markov chain, $X_1, X_2, \dots$, for the CTMC movement process is defined by the transition probabilities

$$q(X_m = j | X_{m-1} = i) = \lambda_{ij} / \Lambda_i \text{ for } i \sim j,$$

where $\Lambda_i = \sum_{j \in \mathcal{N}_i} \lambda_{ij}$. All we need to do is construct an appropriate π_i , $i = 1, \dots, n$, which is only a function of cell i and no other cell j , such that

$$R_m = \frac{\alpha_j \lambda_{ji} \Lambda_i}{\alpha_i \lambda_{ij} \Lambda_j} = 1$$

for all $m > 0$ and all (i, j) such that i and j are neighbors. Then, $\boldsymbol{\pi}$ will be the limiting distribution of the embedded chain and we can use the result that the limiting distribution of the full CTMC is $u_i \propto \pi_i / \Lambda_i$.

We begin by evaluating the MH acceptance ratio for our CTMC model. After canceling like terms in the fraction and rearranging advantageously, we obtain,

$$R_m = \frac{\exp\{-a_i + b_i + 2c_i\} \Lambda_i}{\pi_i} \times \frac{\pi_j}{\exp\{-a_j + b_j + 2c_j\} \Lambda_j} \times \exp\{d_{ji} - d_{ij}\} \quad (1)$$

37 From this MH acceptance ratio, one can see that a sufficient condition to satisfy $R_m = 1$
 38 for all $m = 1, 2, \dots$ and all (i, j) such that $i \sim j$ is that d_{ij} is symmetric. That is, $d_{ij} = d_{ji}$.
 39 Then, $R_m \equiv 1$ for $\pi_i \propto \exp\{-a_i + b_i + 2c_i\}\Lambda_i$. Placing this in to the embedded Markov
 40 chain formula for the limiting distribution of a CTMC we obtain,

$$u_i \propto \frac{\pi_i}{\Lambda_i} = \exp\{-a_i + b_i + 2c_i\}. \quad (2)$$

41 Supplementary Tables and Figures

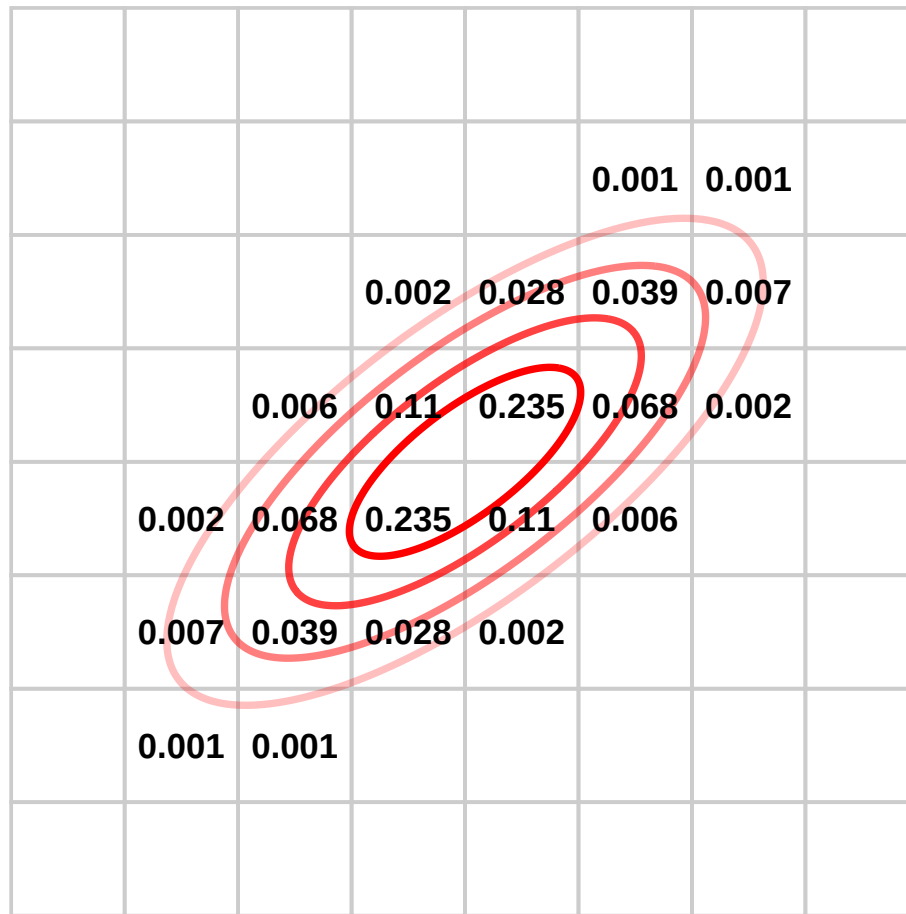


Figure S1: This illustrates the discretization of the continuous-space location error distribution onto the discrete space cells of the study area partition. The red ellipses represent the bivariate normal location distribution and the numbered values are the volume of the bivariate normal destiny function contained within the cells.

Table S1: Tagging information and satellite deployment details.

Tag ID	Catalog ID	Age class at tagging	Sex	Social cluster membership	Tag type	Deployment date (UTC)	Duration (days)
PcTag066	HIPc0202	Adult	Male	Cluster 1	SPOT6	2020-06-22	28.4
PcTag067	HIPc0159	Adult	Female	Cluster 3	SPOT6	2020-09-29	81.1
PcTag070	HIPc0149	Adult	Male	Cluster 2	SPOT6	2020-12-07	42.5
PcTag071	HIPc0711	Adult	Female	Cluster 4	SPOT6	2020-12-12	39.8
PcTag074	HIPc0364	Adult	Unknown	Cluster 3	SPLASH10-F	2021-08-08	12.0

Note: For information on social cluster membership, see Mahaffy et al. (2023). Sex was genetically determined (via biopsy sample) for all individuals except HIPc0364. Transmission durations are rounded to one decimal place.

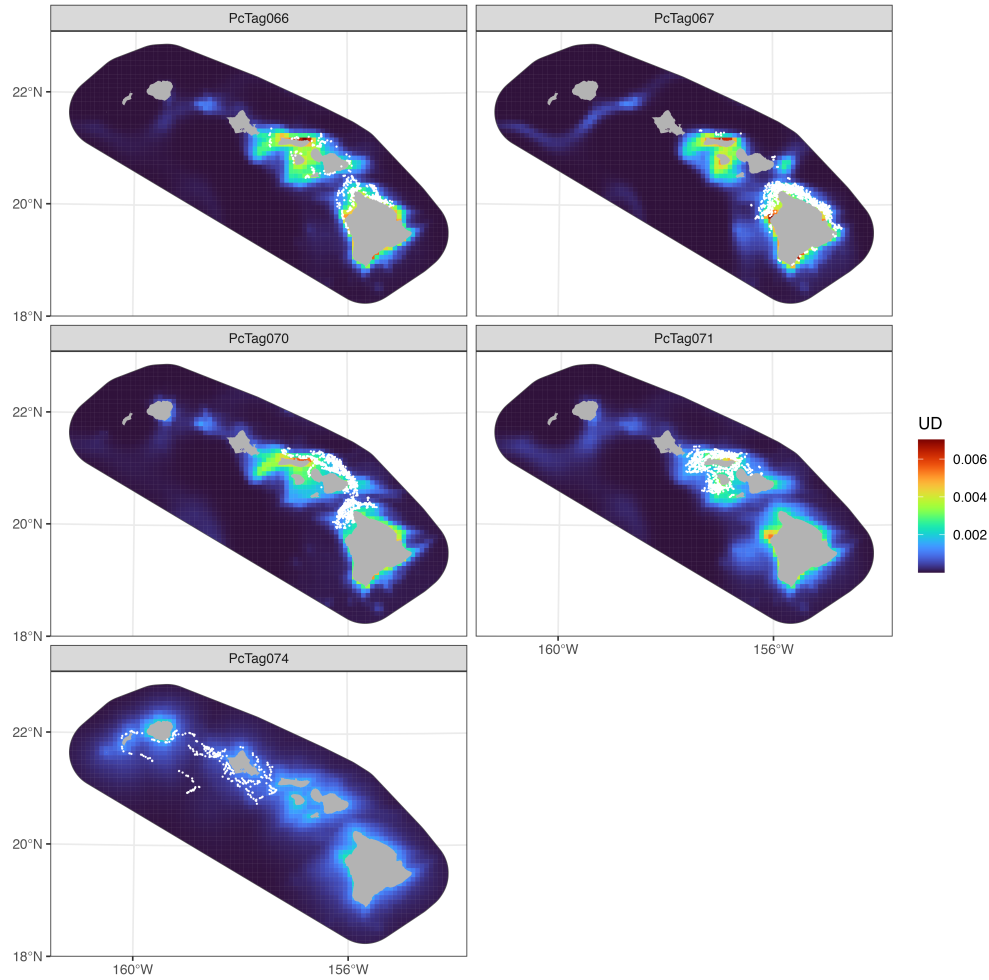


Figure S2: Individual CTMC model predicted utilization distributions. The UD depicted here were constructed from the MLE estimated movement rate coefficients for each individual.

References

- Chib, S. and Greenberg, E. (1995). Understanding the Metropolis-Hastings algorithm. *The American Statistician*, 49(4):327–335.
- Mahaffy, S., Baird, R., Harnish, A., Cullins, T., Stack, S., Currie, J., Bradford, A., Salden, D., and Martien, K. (2023). Identifying social clusters of endangered main Hawaiian Islands false killer whales. *Endangered Species Research*, 51:249–268.